SenNet LW30 / MB30 Meter

Communications via LoRaWAN or Modbus RTU, depending on the project.

The SenNet LW30/MB30 Meter is a DIN-rail three-phase power analyzer designed for integration into modern energy monitoring architectures. Available in two communication versions depending on project requirements: LoRaWAN (EU868 band) for installations without data cabling, or Modbus RTU over RS-485 for direct integration into existing BMS and SCADA platforms.

Class 0.5S Accuracy	Two models with different communications	DIN-rail mounting
Active energy measurement according to IEC 62053-21/22.	LoRaWAN EU868 or Modbus RTU RS-485, depending on the model.	Compact 72 mm format, installation in an electrical panel.

Two models under order

SenNet LW30 Meter	SenNet MB30 Meter
LORAWAN	MODBUS RTU · RS-485
Long-range, low-power wireless connectivity. Ideal when no data cabling is available.	Direct wired integration with existing BMS, SCADA or SenNet controllers in the installation.

The SenNet LW30/MB30 Meter current analyzer measures from 100 A to 5000 A using open-core current transformers, both rigid and flexible.

Part of the SenNet family

Compatible with all datalogger and controller models.

www.satel-iberia.com

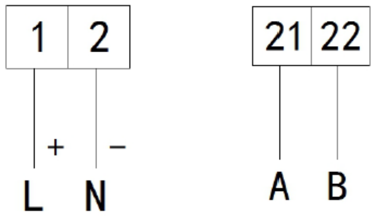
SenNet

Technical specifications

PHYSICAL CHARACTERISTICS	
Dimensions	720x926x715 mm width x height x depth
Display	LCD screen with 4 multifunction buttons
CURRENT INPUT	
Input current	External open-core 333 mV CTs or Rogowski loop-type probe
Wiring	Summing of CTs, in 100 A steps (only for 333 mV probes)
Consumption	< 1 VA (per phase)
FUNCTIONAL CHARACTERISTICS	
Rated voltage	3×57.7/100 V · 3×220/380 V · 3×380/660 V · 3×100 V · 3×380 V · 3×660 V
Consumption	< 0.5 VA (per phase)
Reference frequency	45-65 Hz
Available measurement	V, I, P, Q, S, PF, F, THDv, THDi, harmonics from 2nd to 31st
LoRaWAN module	L-LRNWB25-84DN4
AUXILIARY POWER SUPPLY	
Supply voltage	AC 85 ~ 465 V
Consumption	< 2 W
MEASUREMENT PERFORMANCE	
Standards	IEC 62053-22:2003 · IEC 62053-21:2003
Active energy accuracy	Class 0.5S
PULSE OUTPUT	
Pulse width	80 ± 20 ms
Pulse constant	6400 imp/kWh · 400 imp/kWh
COMMUNICATION	
SenNet LW30 Meter	LoRaWAN EU868 MHz (EU863 – 870) LoRaWAN RU868 MHz LoRaWAN AS923-1/2/3/4 LoRaWAN US915 LoRaWAN AU915 Transmit power: 16 dBm.
SenNet MB30 Meter	Modbus RTU over RS-485 (shielded twisted pair)
ENVIRONMENTAL CONDITIONS	
Operating temperature	-20 °C ~ 60 °C
Storage temperature	-40 °C ~ 70 °C
Relative humidity	≤ 95 % (non-condensing)

Connection diagrams

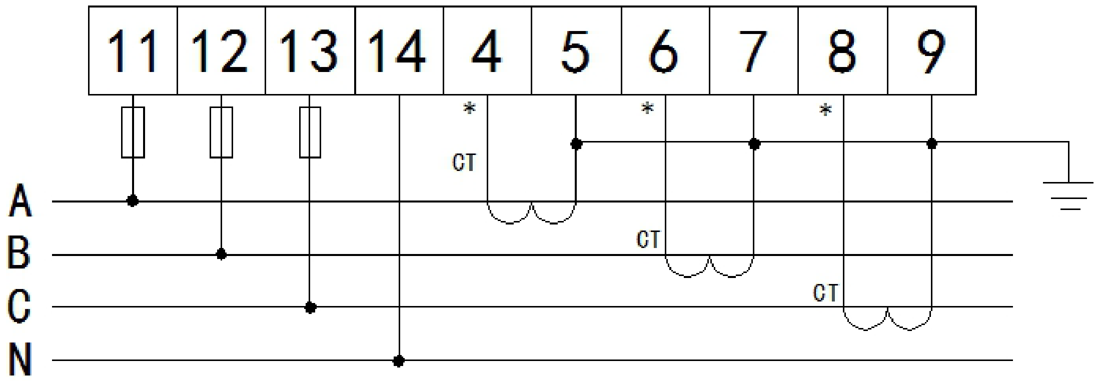
To connect the device, first connect the power supply and data lines in the case of the MB30:



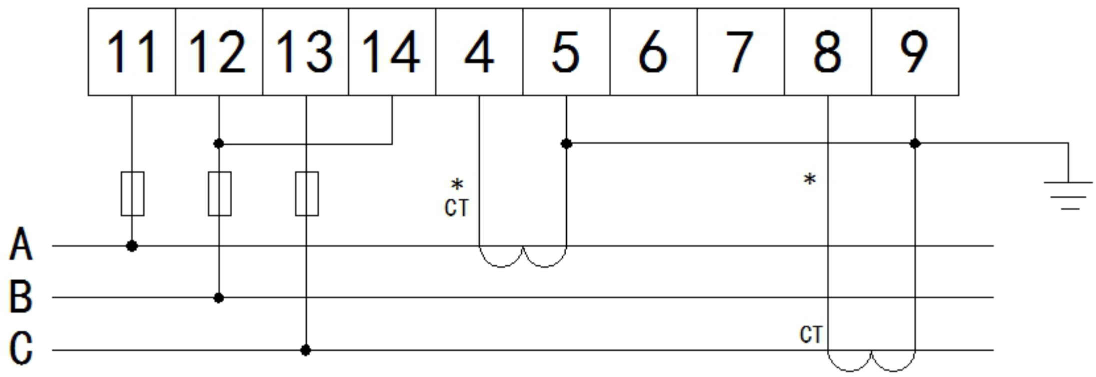
Auxiliary power supply Interface Communication Interface

Then connect the voltage reference and the current transformers. This connection can be made with or without neutral wire, as follows:

- Voltage reference and current transformer wiring for a 4-wire three-phase line:



- Voltage reference and current transformer wiring for a 3-wire three-phase line:



Modbus RTU communications

The following table shows the Modbus addressing of the MB30 analyzer variables for device configuration; these parameters can also be changed using the analyzer display:

Address	Variable	Length	Format	R/W	Note
0	Modbus address	1	Signed	R/W	1-247
1	Speed	1	Signed	R/W	1: 1200 bps 2:2400 bps 3:4800 bps 4:9600 bps
4	Parity	1	Signed	R/W	High byte: 0: none 1: even 2: odd Low byte: 0:1 stop bit 1:2 stop bit
7	Display sleep mode	1	Signed	R/W	Time in seconds
8	Password	1	Signed	R/W	
15	Current probe.	1	Signed	R/W	Multiplier of 100. Example: 1 = 100A 2=200A ... 8=800A
17-19	Time	1	Signed	R/W	Date, time (second, minute, hour, day, month, year)

The following table shows the Modbus addressing of the MB30 analyzer variables for reading parameters:

Address	Variable	Length	Format	R/W	Unit
32768	Phase 1 voltage	2	Float	R	V
32770	Phase 2 voltage	2	Float	R	
32772	Phase 3 voltage	2	Float	R	
32774	Phase 1-2 voltage	2	Float	R	
32776	Phase 2-3 voltage	2	Float	R	
32778	Phase 3-1 voltage	2	Float	R	
32780	Phase 1 current	2	Float	R	A
32782	Phase 2 current	2	Float	R	
32784	Phase 3 current	2	Float	R	
32788	Phase 1 active power	2	Float	R	kW
32790	Phase 2 active power	2	Float	R	
32792	Phase 3 active power	2	Float	R	
32794	Total active power	2	Float	R	
32796	Phase 1 reactive power	2	Float	R	kVar
32798	Phase 2 reactive power	2	Float	R	
32800	Phase 3 reactive power	2	Float	R	
32802	Total reactive power	2	Float	R	
32804	Phase 1 apparent power	2	Float	R	kVA
32806	Phase 2 apparent power	2	Float	R	
32808	Phase 3 apparent power	2	Float	R	
32810	Total apparent power	2	Float	R	
32812	Phase 1 power factor	2	Float	R	
32814	Phase 2 power factor	2	Float	R	
32816	Phase 3 power factor	2	Float	R	
32818	Total power factor	2	Float	R	
32820	Frequency	2	Float	R	Hz
34942	Total active energy	2	Float	R	kWh
34944	Active energy consumed	2	Float	R	
34946	Active energy generated	2	Float	R	
34948	Total reactive energy	2	Float	R	kVArh
34950	Reactive energy consumed	2	Float	R	
34952	Reactive energy generated	2	Float	R	

The following table shows the Modbus addressing of the MB30 analyzer variables for reading total and individual harmonics from 2nd to 31st:

Address	Variable	Length	Format	R/W	Factor
116	THDV phase 1	1	Unsigned	R	
117	THDV phase 2	1	Unsigned	R	
118	THDV phase 3	1	Unsigned	R	
119	THDI phase 1	1	Unsigned	R	
120	THDI phase 2	1	Unsigned	R	
121	THDI phase 3	1	Unsigned	R	
122-151	Phase 1 voltage harmonic, 2nd to 31st	1	Unsigned	R	0.01
152-181	Phase 2 voltage harmonic, 2nd to 31st	1	Unsigned	R	
182-211	Phase 3 voltage harmonic, 2nd to 31st	1	Unsigned	R	
212-241	Phase 1 current harmonic, 2nd to 31st	1	Unsigned	R	
242-271	Phase 2 current harmonic, 2nd to 31st	1	Unsigned	R	
272-301	Phase 3 current harmonic, 2nd to 31st	1	Unsigned	R	

LoRaWAN communications

The following table shows the LoRaWAN addressing of the LW30 analyzer variables for device configuration; these parameters can also be changed using the analyzer display:

Address	Variable	Length	Format	R/W	Note
0	Modbus address	1	Signed	R/W	1-247
1	Speed	1	Signed	R/W	1: 1200 bps 2:2400 bps 3:4800 bps 4:9600 bps
2	Spreading factor	1	Signed	R/W	6-12
3	Operating frequency	1	Signed	R/W	0-45
4	Parity	1	Signed	R/W	High byte: 0: none 1: even 2: odd Low byte: 0:1 stop bit 1:2 stop bit
7	Display sleep mode	1	Signed	R/W	Time in seconds
8	Password	1	Signed	R/W	
15	Current probe.	1	Signed	R/W	Multiplier of 100. Example: 1 = 100A 2=200A ... 8=800A
17-19	Time	1	Signed	R/W	Date, time (second, minute, hour, day, month, year)

The following table shows the Modbus addressing of the MB30 analyzer variables for reading parameters:

Address	Variable	Length	Format	R/W	Unit
32768	Phase 1 voltage	2	Float	R	
32770	Phase 2 voltage	2	Float	R	
32772	Phase 3 voltage	2	Float	R	V
32774	Phase 1-2 voltage	2	Float	R	
32776	Phase 2-3 voltage	2	Float	R	
32778	Phase 3-1 voltage	2	Float	R	
32780	Phase 1 current	2	Float	R	
32782	Phase 2 current	2	Float	R	A
32784	Phase 3 current	2	Float	R	
32788	Phase 1 active power	2	Float	R	
32790	Phase 2 active power	2	Float	R	kW
32792	Phase 3 active power	2	Float	R	
32794	Total active power	2	Float	R	
32796	Phase 1 reactive power	2	Float	R	
32798	Phase 2 reactive power	2	Float	R	kVar
32800	Phase 3 reactive power	2	Float	R	
32802	Total reactive power	2	Float	R	
32804	Phase 1 apparent power	2	Float	R	
32806	Phase 2 apparent power	2	Float	R	kVA
32808	Phase 3 apparent power	2	Float	R	
32810	Total apparent power	2	Float	R	
32812	Phase 1 power factor	2	Float	R	
32814	Phase 2 power factor	2	Float	R	

32816	Phase 3 power factor	2	Float	R	
32818	Total power factor	2	Float	R	
32820	Frequency	2	Float	R	Hz
34942	Total active energy	2	Float	R	kWh
34944	Active energy consumed	2	Float	R	
34946	Active energy generated	2	Float	R	
34948	Total reactive energy	2	Float	R	kVArh
34950	Reactive energy consumed	2	Float	R	
34952	Reactive energy generated	2	Float	R	

For more technical information, additional datasheets or integration support, visit **www.satel-iberia.com**